Ex.no: 3

Date:

Experiments on CISCO PACKET TRACER

a.) Experiment:

Understanding the cisco packet Environment

- Aim:

To understanding the cisco packet tracker environment and design a simple computer network using its interface and networking components

- Steps:

1. Open the Cisco packet Tracker application.
2. Observe the user interface, including, common tools Bar, Main Toolbar, workspace, common tools Bar, Network Component Box and Realtime / simulation tabs
3. From the network components Box, select the Required network devices such as PCs, Hub and Switch.
4. Drag and drop the devices into the workspace.
5. Connect the devices using copper straight through cables.

6. Configure IP addresses after for the PC's using the Desktop -> IP. Configuration option.

7. Verify that all links are active.

8. Observe packet transmission in Realtime mode and simulator mode.

9. Save the network topology.

* Result!

The Cisco packet tracer environment was successfully explored and simple network was designed.

1.) HUB

* Experiment 1 : HUB

* Aim :

To study the behaviour of a Hub by observing how it forwards data packets using Cisco Packet tracer.

* Steps :

1. Open Cisco Packet Tracer.
2. Drag and place 1 Hub and 4 generic PCs into the workspace.
3. Connect all PCs to the HUB using Copper straight through cables.
4. Configure IP addresses for all PC's in the same network.
5. Select Add Simple PDU from the common Tools bar.
6. Send a packet from one PC to another PC
7. Switch to Simulation Mode and observe how the Hub forwards the Packet.

* Result :

The Hub broadcast the received packet to all connected devices. Every PC receives the packet, but only destination PC accepts it, while the others discard it.

2.) SWITCH

. Experiment 2 : SWITCH

* Aim:

To study the behavior of a switch by observing how it forwarded data packets using Cisco Packet Tracer.

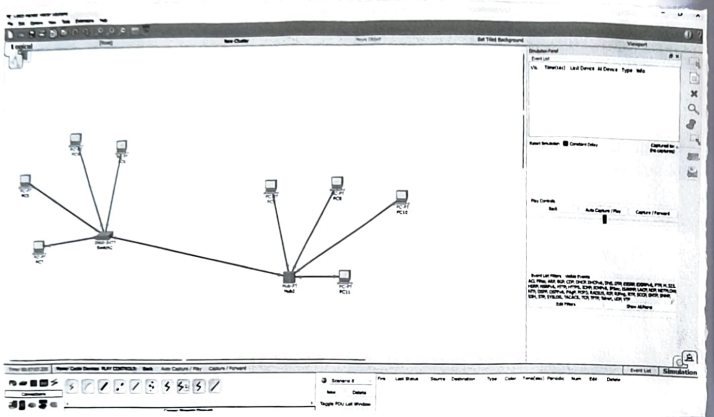
* Steps:

1. Open Cisco Packet Tracer.
2. Drag and place 1 Switch and 4 Generic PCs into the workspace.
3. Connect all PCs to the Switch using Copper straight-through cables.
4. Configure IP addresses for all PCs in the same network.
5. Select Add simple PDU from the common Tools bar.
6. Send a packet from one PC to another PC.
7. Switch to Simulation mode and observe how the switch forwards the packet.

* Result:

The Switch forward the packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency.

3) HUB AND SWITCH.



. Experiment 3: HUB and SWITCH (comparision)

* Aim:

To compare the packet forwarding behavior of a HUB and a Switch using Cisco Packet Tracer.

* Steps:

1. Create one network using a Hub and another using a Switch, each connected to 4 PCs.
2. Configure IP addresses for all PC's in both networks.
3. Send a Simple PDU between two PCs in each network.
4. Observe packet transmission in Simulation mode.
5. Compare how the HUB and Switch forward packets.

* Result:

The Hub broadcasts packet to all connected devices, whereas the Switch forwards packets only to the intended destination. Therefore, the Switch provides better performance, reduced network traffic and improved efficiency compared to the Hub.